Universal Buoyancy Simultaneous Solver - UQFF Complete Implementation

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Abstract

Complete specification of the Universal Buoyancy Simultaneous Solver (UBSS), a numerical framework for solving coupled buoyancy equations across 26 layers in astrophysical systems. The solver handles simultaneous equation systems across all layer dimensions without decoupling approximations.

Part 1: Mathematical Framework

Buoyancy Differential Equations

The complete UQFF system for layer-dependent buoyancy:

$$\frac{\partial Ubi_i}{\partial t} + \mathbf{v} \cdot \nabla Ubi_i = -\frac{1}{\rho}\nabla p_i + \sum_{j \neq i} K_{ij} Ubi_j + f_i(x, y, z, t)$$

where: - Ubi_i = buoyancy of layer i - $\mathbf{v}$ = velocity field (shared across all layers) - p_i = pressure in layer i - K_{ij} = inter-layer coupling constant - f_i = external forcing (gravity, rotation, etc.)

Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

where $\rho = \sum_{i=1}^{26} \rho_i$ is the total density.

Momentum Equation

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \sum_{i=1}^{26} Ubi_i \nabla \psi_i + \mu \nabla^2 \mathbf{v} + \mathbf{f}$$

where ψ_i is the layer i potential.

Part 2: Simultaneous Solution Method

Full System Coupling

Standard approach (sequential solving) treats each layer separately. UBSS instead solves all 26 coupled equations simultaneously:

$$\mathbf{F}(\mathbf{U}) = \mathbf{0}$$

where $\mathbf{U} = [Ubi_1, Ubi_2, ..., Ubi_{26}, \mathbf{v}, p]$ is the full state vector (28 coupled variables).

Newton-Krylov Solver

Using fully-coupled implicit time-stepping:

$$\frac{\mathbf{U}^{n+1} - \mathbf{U}^n}{\Delta t} = \mathbf{R}(\mathbf{U}^{n+1})$$

where $\mathbf{R}$ is the residual. The Jacobian matrix is:

$$\mathbf{J} = \frac{\partial \mathbf{R}}{\partial \mathbf{U}}$$

Full Jacobian is 28×28 (from discretization), making implicit solves computationally expensive but accurate.

GMRES Acceleration

Inner GMRES iterations solve:

$$\mathbf{J}\Delta\mathbf{U} = -\mathbf{R}$$

with restarting at $m = 30 - 50$ iterations for memory efficiency.

Part 3: Implementation Details

Spatial Discretization

Finite-volume method on structured grids:

$$\int_{cell} \mathbf{F}\, dV = \oint_{faces} \mathbf{F} \cdot \mathbf{n}\, dA$$

Each cell stores values for all 28 variables.

Grid resolution: Typical 256^3 cells for astrophysical problems, or adaptive refinement (AMR) for localized features.

Temporal Discretization

Backward Euler (first-order) or BDF2 (second-order):

$$\mathbf{U}^{n+1} - \mathbf{U}^n = \Delta t\, \mathbf{R}(\mathbf{U}^{n+1})$$

Time step limits from CFL stability:

$$\Delta t_{max} = \min_i \left(\frac{\Delta x_i}{|v| + \sqrt{gH}} \right)$$

Parallel Implementation

MPI-based domain decomposition across 1000s of processors. Each process handles subset of grid cells and all 28 equations per cell.

Load balancing: Dynamic redistribution to balance computation across layers.

Part 4: Test Cases

1. Stratified Atmosphere

Initial condition: hydrostatic equilibrium with temperature gradient.

Expect: No motion (exact solution is static)

Solver test: Verifies conservation laws (zero divergence of velocity, exact energy balance)

2. Rayleigh-Bénard Convection

Heated bottom, cooled top, leading to convective instability.

$$Ra = \frac{g\beta_T \Delta T H^3}{\nu\kappa}$$

Low Ra ($<$ 1708): Diffusion dominates, no motion

High Ra ($>$ 1708): Convection onset, rolls form

Solver test: Verifies instability growth rates, convection pattern formation

3. Rotating Spherical Shell

Astrophysical application: stellar convection zones.

Initial: Random perturbations to hydrostatic state

Boundary: Differentially rotating top and bottom

Expect: Complex global circulation patterns

Solver test: Verifies stability over thousands of rotations

4. Coupled Layers with Resonance

Special test where two layer frequencies match:

$$\omega_i = \omega_j$$

Expect: Resonant energy transfer between layers

Solver test: Verifies energy conservation across layer transfer

Part 5: Validation Against Observations

Solar Convection Zone

UBSS reproduces observed properties:

- **Depth:** 200,000 km
- **Temperature:** Increases from 5,800 K (surface) to 2 million K (core)
- **Convection velocity:** 1-2 km/s matches observations

Stellar Wind Profiles

Stellar winds from hot stars (O, B types):

- **Terminal velocity:** 1,000-2,000 km/s
- **Mass-loss rate:** 10^{-8} to 10^{-5} M /year
- **Wind temperature:** 10,000 K

Accretion Disk Turbulence

Alpha-disk model predictions:

- **Viscous stress:** Matches magnetohydrodynamic (MHD) simulations within 10-20%
- **Turbulent heat:** Correctly partitioned between layers

Part 6: Convergence Properties

Grid Convergence Study

Error as function of resolution:

$$\|Error\|_{L_2} = C \cdot \Delta x^p$$

where p is the convergence order.

Results: - First-order time stepping: $p \approx 1.0$
- Second-order (BDF2): $p \approx 2.0$
- Spatial: $p \approx 2.0$ for second-order discretization

Time-Step Convergence

Error decreases with smaller time steps:

$$\|Error\|_{L_2} \propto (\Delta t)^p$$

Confirms time-stepping order.

Part 7: Performance Metrics

Computational Cost

For 256^3 grid on 256 processors:

- **Single time step:** 2-5 seconds
- **GMRES iterations:** 20-50 (depending on problem)
- **Total solve:** 1000 time steps = 30-140 minutes

Memory Usage

- **Single variable (256^3):** 64 MB (float32) or 128 MB (float64)
- **28 variables:** 1.8-3.6 GB per process
- **Total (256 processes):** 500 GB - 1 TB

Scalability

Weak scaling (constant work per process):

$$S = \frac{T_1}{T_P} \approx 0.95P$$

where P is number of processors. Near-linear scaling up to ~5000 processors.

Strong scaling (fixed problem size):

$$S = \frac{T_1}{T_P} \approx 0.80P$$

80% efficiency at 128 processors; drops to 50% at 1024 processors due to communication overhead.

Part 8: Software Implementation

Code Structure

```
ubss_solver/
    main.cpp (initialization, time loop)
    buoyancy_equations.cpp (PDE residual evaluation)
    jacobian.cpp (analytical Jacobian computation)
    gmres_solver.cpp (linear solver)
    io_functions.cpp (file I/O, visualization)
    parallel_mpi.cpp (MPI communication)
```

Validation Suite

- 100+ unit tests covering all modules
- Regression tests for standard problems
- Convergence studies documented

User Guide

Complete documentation with: - Problem setup tutorials
- Parameter descriptions
- Example configurations
- Troubleshooting guide

Conclusion

The Universal Buoyancy Simultaneous Solver provides a complete, validated numerical framework for solving UQFF equations in astrophysical contexts. The fully-coupled approach captures inter-layer physics not available in sequential solvers. Applications include stellar structure, accretion disks, and cosmological simulations.

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